1. a) Figure 1.1 below illustrates an RLC circuit with the standard interpretation of symbols. The input is $v_i(t)$ and the output is $q(t)$, the capacitor charge.

i) Derive the differential equation relating q to v_i. [5]

ii) Determine the transfer function relating q to v_i. [5]

iii) Let $L = 1\ H$ and let $v_i(t)$ be a unit step input. The following design specifications are required to be satisfied (S1): The capacitor charges to its steady–state value within approximately 10^{-3} seconds. (S2): The maximum overshoot of $q(t)$ is 5% of its steady–state value.

A. Derive the values of R and C so that the design specifications are satisfied. [5]

B. For these values of R and C, derive the steady state value of $q(t)$. [5]

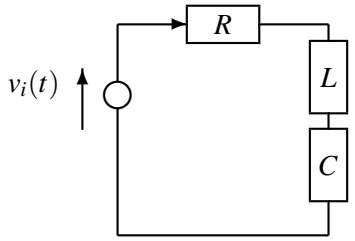


Figure 1.1

b) Figure 1.2 below illustrates an aircraft's autopilot where K_1 and K_2 are design parameters.

i) Derive the transfer function that relates the error signal to the reference signal. [5]

ii) Use the Routh-Hurwitz criterion to find the maximum value of K_1 (in terms of K_2) for closed–loop stability. [5]

iii) Calculate the steady-state error (in terms of K_1 and K_2) when the reference is a unit ramp function. [5]

iv) Let $K_2 = 1$. Use the answers above to find the minimum value of the achievable steady–state error when the reference is a unit ramp. [5]

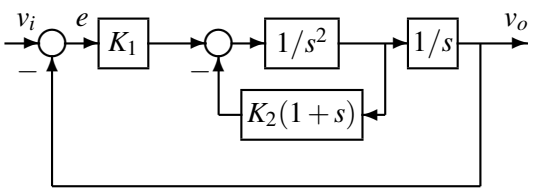


Figure 1.2

2. Consider the feedback control system in Figure 2.1 below.

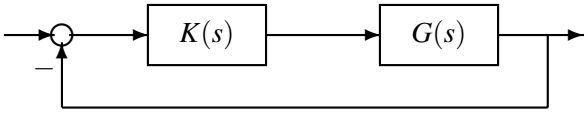


Figure 2.1

Here, $K(s)$ is the transfer function of a compensator while $G(s)$ is a stable transfer function with no finite zeros whose frequency response is shown in Figure 2.2.

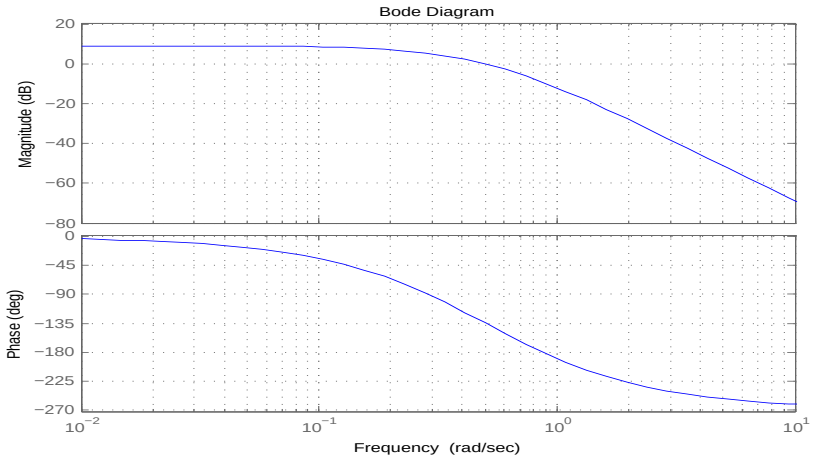


Figure 2.2

a) Use the frequency response to sketch a **rough** Nyquist diagram of $G(s)$, indicating the low and high frequency portions and the real–axis intercepts. [8]

b) Give approximate values for the gain and phase margins. [8]

c) Use the Nyquist stability criterion, which should be stated, to determine the number of unstable closed-loop poles when:

i) $K(s) = 1$, [4]

ii) $K(s) = 10$. [4]

d) Let $K(s)$ have the frequency response shown in Figure 2.3 overleaf. Describe $K(s)$ briefly and indicate its effects on the performance and stability of the feedback loop. [6]

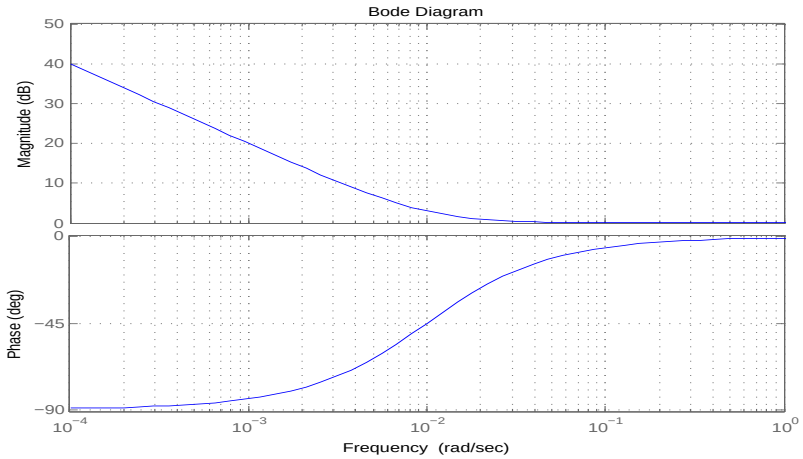

Bode Diagram
Magnitude (dB)
50
40
30
20
10
0
Phase (deg)
0
−45
−90
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Frequency (rad/sec)

Figure 2.3

3. Consider the feedback loop shown in Figure 3.1 below. Here

$$G(s) = \frac{1}{(s+2)^2}.$$

It is required design a constant compensator k such that the following design specifications are satisfied:

- The settling time is approximately 2 seconds.
- The response is oscillatory with a maximum overshoot of approximately 5%.

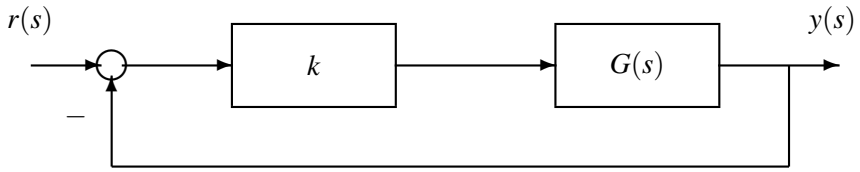


Figure 3.1

a) Find the location of the closed–loop poles that achieves the design specifications above. [5]

b) Write down the closed–loop characteristic equation in terms of k. [5]

c) Use the Routh–Hurwitz criterion to determine the range of values of k for which the closed–loop is stable. [6]

d) Find the value of k that achieves the design specifications. [8]

e) For this value of k, use the final value theorem to find the steady–state error when $r(t)$ is a unit step. [6]

SOLUTIONS: Feedback Systems

1. *The students have done relatively well on this question, scoring approximately 65%.*

a) *This is an RLC circuit modeling and design question and is a variation on typical study group questions.*

i) Applying Kirchhoff's law on the loop,

$$\boxed{v_i(t) = L\ddot{q}(t) + R\dot{q}(t) + C^{-1}q(t).}$$

ii) Taking Laplace transform gives the transfer function

$$\boxed{\frac{q(s)}{v_i(s)} = \frac{1}{Ls^2 + Rs + C^{-1}}}$$

iii) Comparing the transfer function with the standard second order form

$$G(s) = C\frac{(LC)^{-1}}{s^2 + sRL^{-1} + (LC)^{-1}} = K\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

gives $K = C$, $\omega_n = \frac{1}{\sqrt{LC}}$ and $\zeta = 0.5R\sqrt{\frac{C}{L}}$. The second specification demands $\zeta = \frac{1}{\sqrt{2}}$ for 5% maximum overshoot while the first demands $\frac{4}{\zeta\omega_n} = 10^{-3}$.

A. It follows that $\boxed{R = 8 \times 10^3\ \Omega}$ and $\boxed{C = 31.25 \times 10^{-9}\ F}$.

B. The steady state output is simply $G(0) = C$ and so $\boxed{q_{ss} = 31.25 \times 10^{-9}.}$

b) i) A computation gives

$$\boxed{\frac{e(s)}{v_i(s)} = \frac{s(s^2 + K_2 s + K_2)}{s^3 + K_2 s^2 + K_2 s + K_1}}$$

ii) The Routh array is given by

s^3	1	K_2
s^2	K_2	K_1
s	$\frac{K_2^2 - K_1}{K_2}$	
1	K_1	

It follows that $\boxed{K_2 > 0,\ K_1 > 0 \text{ and } K_1 < K_2^2}$ for closed–loop stability.

iii) For a ramp, $v_i(s) = 1/s^2$. Using the final value theorem:

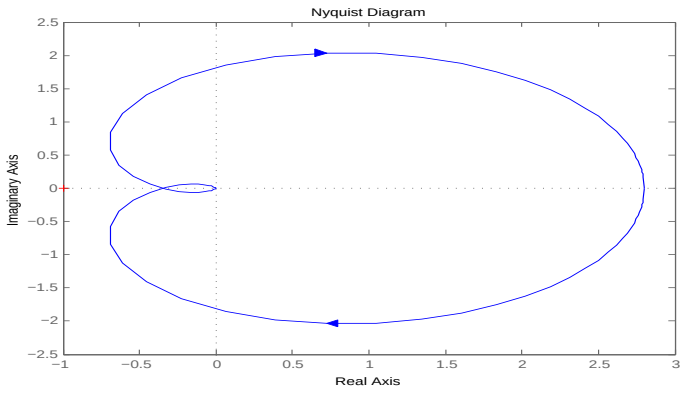
$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} se(s) = \lim_{s\to 0} s\frac{1}{s^2}\frac{s(s^2 + K_2 s + K_2)}{s^3 + K_2 s^2 + K_2 s + K_1} = \boxed{\frac{K_2}{K_1}}$$

iv) Since $K_2 = 1$, $K_1 < 1$ for stability and the steady–state error is $1/K_1$. It follows that the minimum value of the steady–state error is $\boxed{1}$

2. *This question requires obtaining Nyquist diagrams from frequency plots and carrying out analysis and design. The students did very well on this question scoring approximately 75%.*

The transfer function used in fact was $G(s) = \frac{0.35}{(s+0.5)^3}$, although this is not required.

a) The real axis intercepts can be obtained from the frequency response (when the phase is $0, -180^\circ$ and -270° and are approximately given by $\boxed{2.8, -0.35 \text{ and } 0.}$ The Nyquist plot is given below.



b) From the intercepts above, the gain margin is approximately $\boxed{2.9}$. The phase margin can be obtained from the frequency response (by inspecting the phase when the gain is 1) and is approximately $\boxed{45^\circ}$.

c) Let $K(s) = k$. The Nyquist criterion states that $N = Z - P$, where N is the number of clockwise encirclements by the Nyquist diagram of $-k^{-1}$, P is the number of unstable open–loop poles and Z is the number of unstable closed–loop poles. Since $G(s)$ is stable, $P = 0$. An inspection of the Nyquist diagram shows that

i) When $k = 1$, $N = 0$ so $\boxed{Z = 0}$

ii) When $k = 10$, $N = 2$ so $\boxed{Z = 2}$

d) An inspection of the frequency response reveals this is a proportional–plus–integral (PI) compensator. This can be written as

$$K(s) = K_P + \frac{K_I}{s} = \frac{K_I}{s}(1 + \frac{s}{K_I/K_P})$$

It has high gain at frequencies below $\omega_0 = K_I/K_P$ and gain close to K_P beyond ω_0. The phase is negative and large below ω_0 but insignificant above. It follows that by varying K_I and K_P we can use PI compensation to increase low frequency gain (hence improving tracking properties) without introducing phase-lag at high frequency (which would reduce the phase margin) by placing w_0 in the 'middle' frequency range. Since the cross–over frequency for $G(s)$ is approximately 0.8 and ω_0 for $K(s)$ is approximately 0.1, this condition is satisfied.

3. *This is a question about control system design using pole placement and the characteristic equation. The students did relatively well on this question, scoring approximately 73%.*

a) For a maximum overshoot of 5% and a settling time of 2 seconds the closed–loop poles must be placed at $s_1,\ \bar{s}_1 = -2 \pm j2$.

b) The closed–loop characteristic equation is $1 + kG(s) = 0$ or

$$s^2 + 4s + 4 + k = 0.$$

c) The Routh array is given by

$$\begin{array}{c|cc} s^3 & 1 & 4+k \\ s^2 & 4 & 0 \\ s & 4+k & \end{array}$$

Thus for closed–loop stability, $4 + k > 0$ or $\boxed{k > -4.}$

d) To achieve the design specifications, $s_1 = -2 + j2$ and $\bar{s}_1$ must be roots of the characteristic equation. It follows that we must have

$$s^2 + 4s + 4 + k = (s - s_1)(s - \bar{s}_1) = s^2 + 4s + 8.$$

Therefore $\boxed{k = 4.}$

e) Since $r(t)$ is a unit step, $r(s) = 1/s$. Let $e(s) = r(s) - y(s)$. Then

$$e(s) = \frac{r(s)}{1 + kG(s)} = \frac{1/s}{1 + 4/(s+2)^2}$$

The final value theorem then gives

$$\lim_{t \to \infty} e(t) = \lim_{s \to 0} se(s) = \lim_{s \to 0} \frac{1}{1 + 4/(s+2)^2}$$

Therefore $\boxed{e_{ss} = 0.5.}$